

Beethoven Clock — Programming Guide

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Mk. I

Purpose

This programming guide details the steps required to program the firmware on Beethoven Clock's Brain PCB. It also details steps to program the audio file onto the onboard memory.

Required Materials

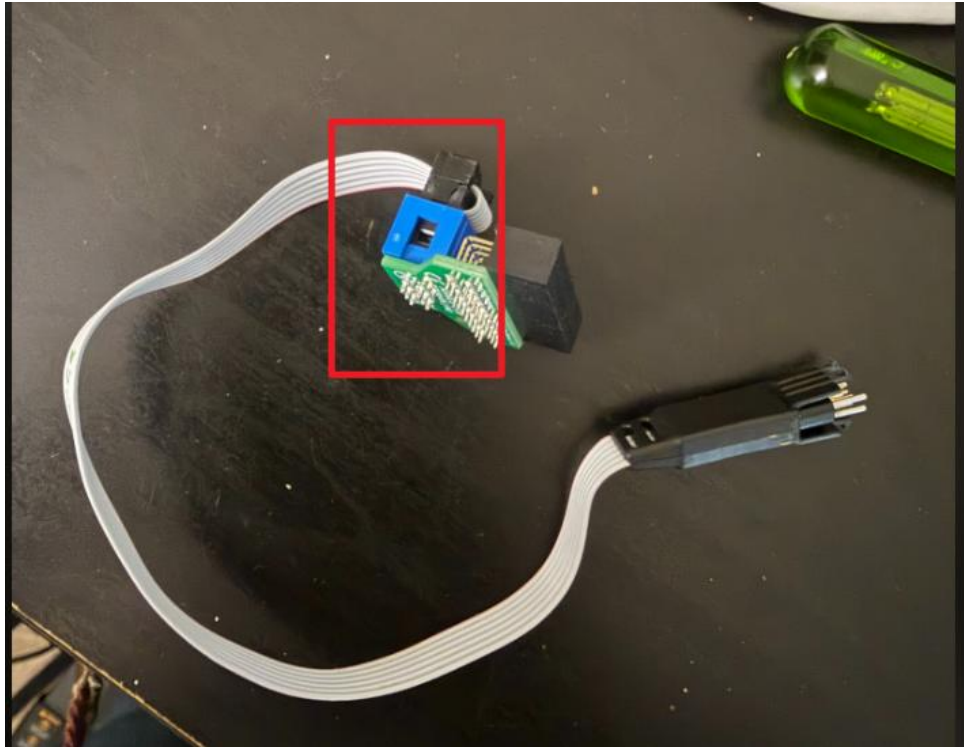
Item Description	Item Manufacturer	Manufacturer Part Number
ST LINK V2	STMicroelectronics	ST-LINK/V2
6-Pin Tag-Connect Plug-of-Nails™ Spring-Pin Cable with Legs	Tag Connect	TC2030-IDC
20-Pin to TC2030-IDC Adapter for Cortex	Tag Connect	ARM20-CTX
FLASHCATUSB Classic Memory Programmer	EmbeddedComputers	FlashcatUSB Classic
8-Pin SOIC Clip	Pomona Electronics**	5250**

**I've been using this 8-SOIC clip, but you can use any similar clip that will work.

Programming Steps

1. Connect the Programming Cable/Adapter to the STLink

Connect the programming cable to the adapter board:

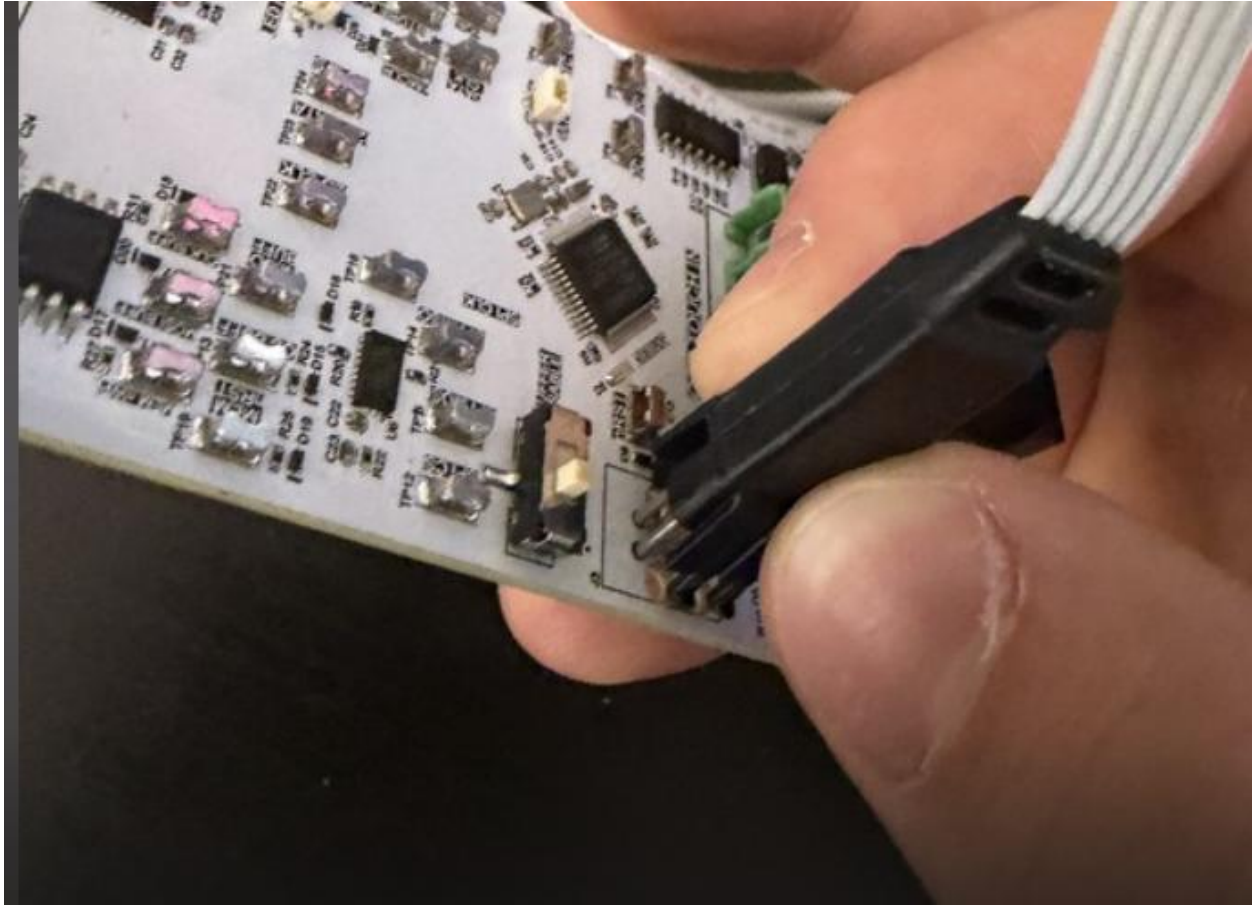


Then connect the adapter board to the STLink's 20-pin header:

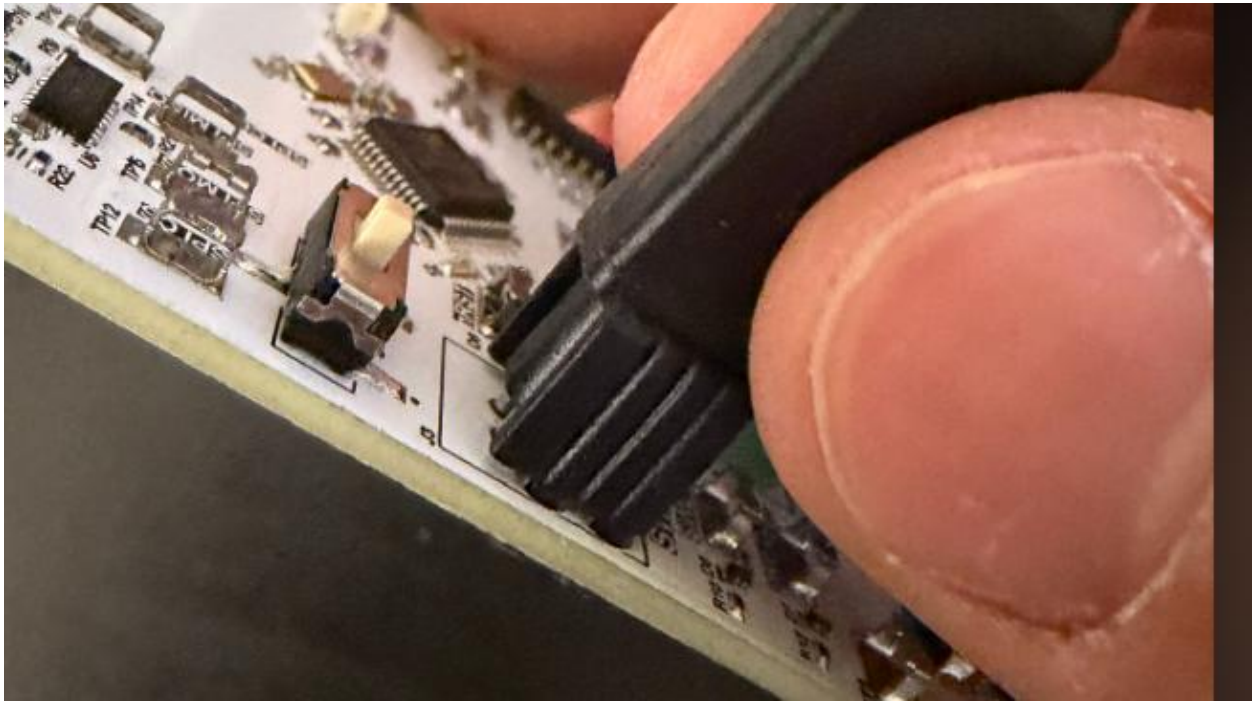


2. Connect the Programming Cable/Adapter to the Beethoven Clock Brain PCB

Line up the little holes in the SWD footprint on the Beethoven Clock Brain PCB with the silver metal poles on the programming cable. Squeeze the 4 black plastic legs that shroud the metal pins:



While squeezing the plastic legs, press the header into the footprint. When you release squeezing, the cable should be securely fastened to the PCB.

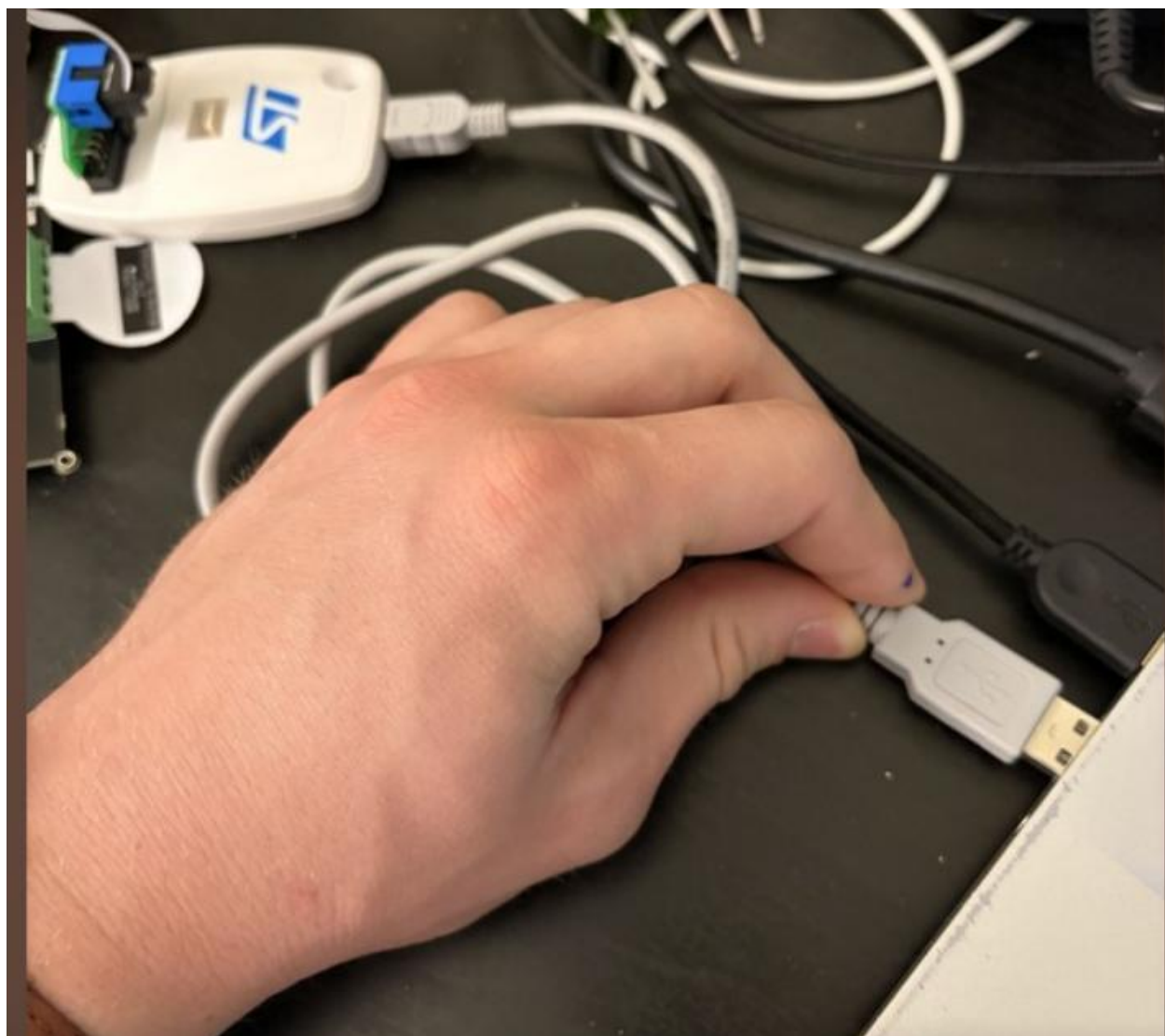


3. Power the Brain PCB on with the Beethoven Clock's Barrel Jack Power Cable

Turn the Brain PCB on. If the Brain PCB is not powered, the microcontroller chip will not be able to be programmed.

4. Connect the STLink to your computer

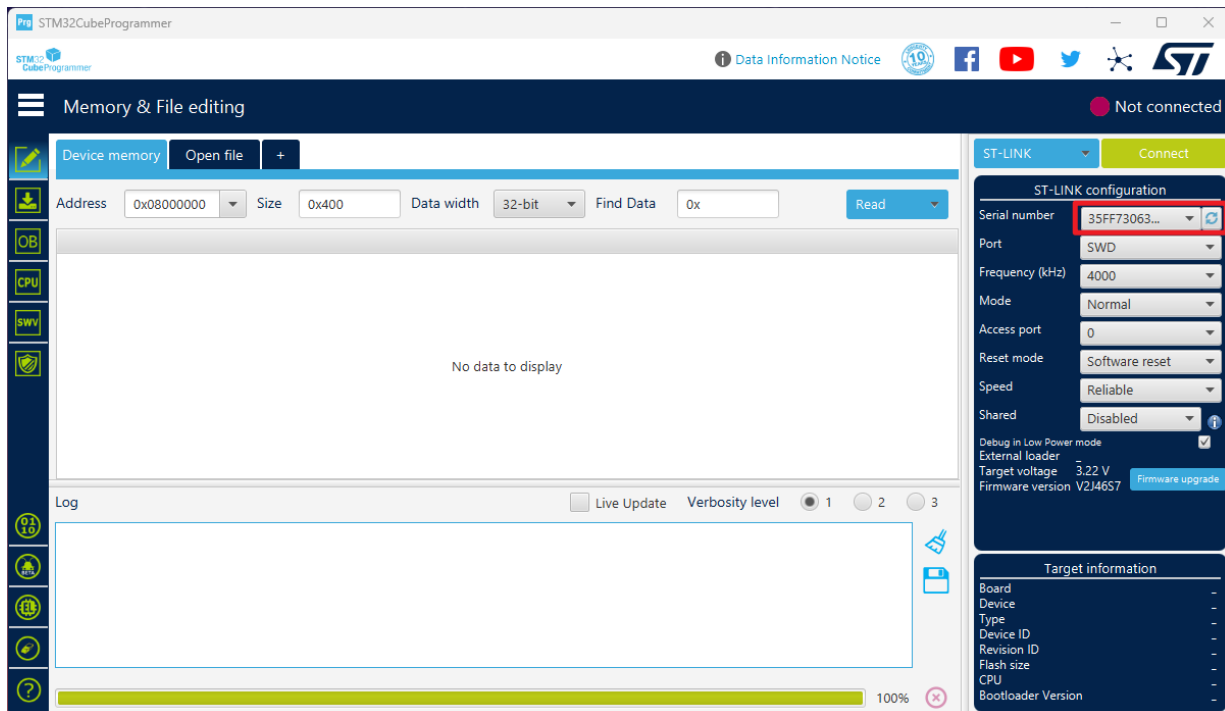
Plug the STLink's USB cable into your computer.



5. Open STM32CubeProgrammer

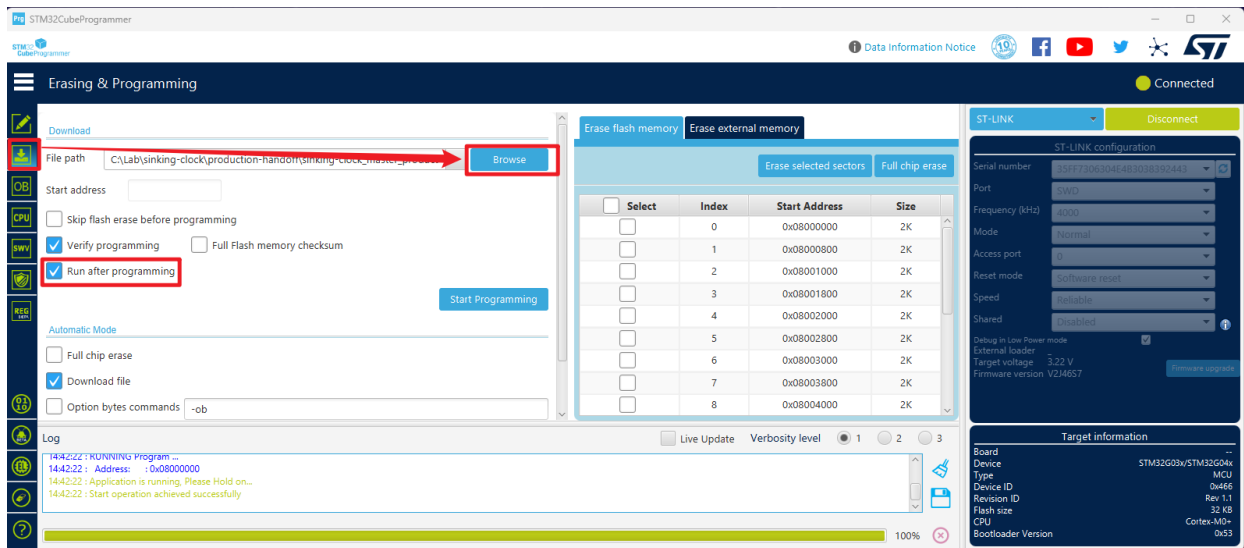
First time doing this: Download the software from [here](#). I believe you use a Mac, so be sure to download the Mac version.

Open STM32CubeProgrammer. Since you already plugged the STLink in, your computer should be able to find the STLink (boxed red):

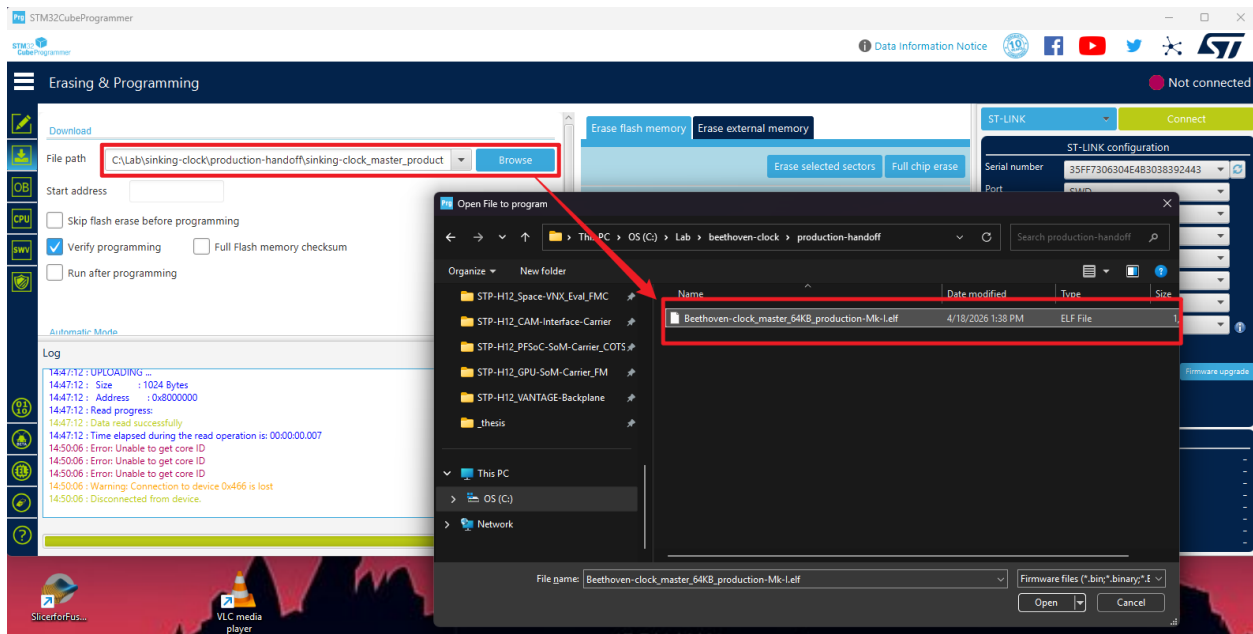


6. Tell STM32CubeProgrammer where the firmware is

Click the down-arrow icon on the left. The STLink will start blinking. Then click “Browse.” You can also check “Run after programming” to make the code run immediately, else you will have to power cycle the board before the code runs correctly.

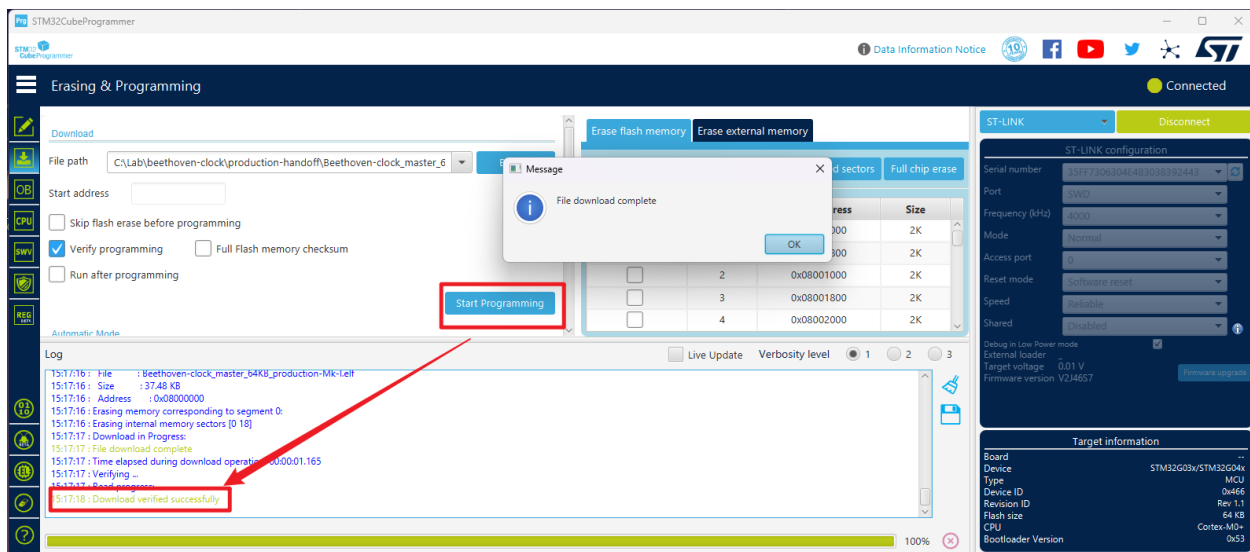


Click the most up-to-date firmware file:

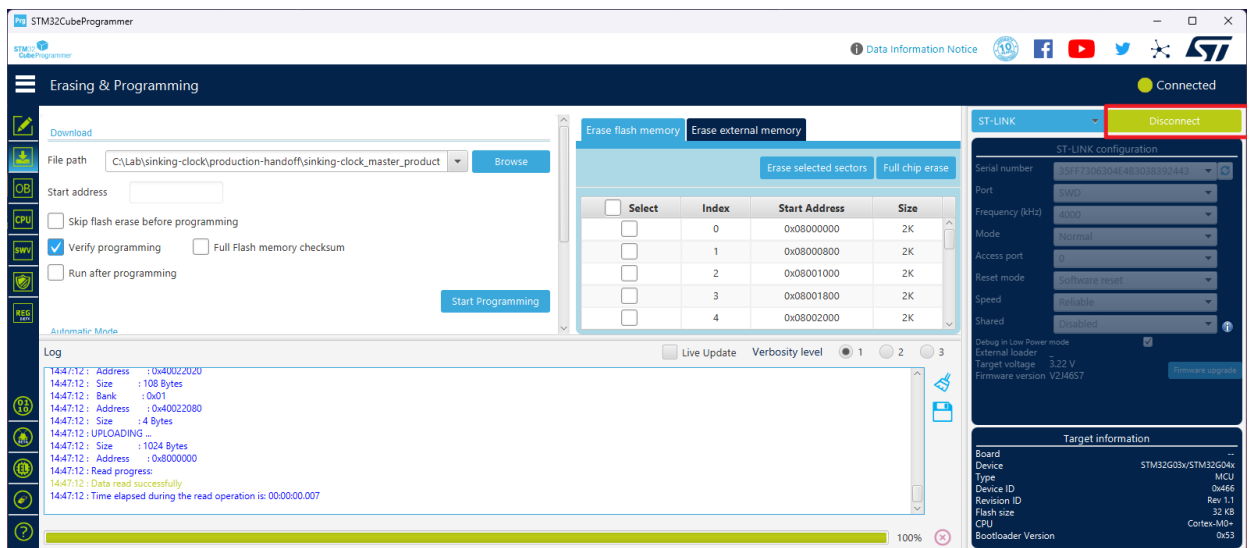


7. Program the Controller chip

Click “Start Programming” and wait until the “Download Verified Successfully” message pops up.



Click “Disconnect”



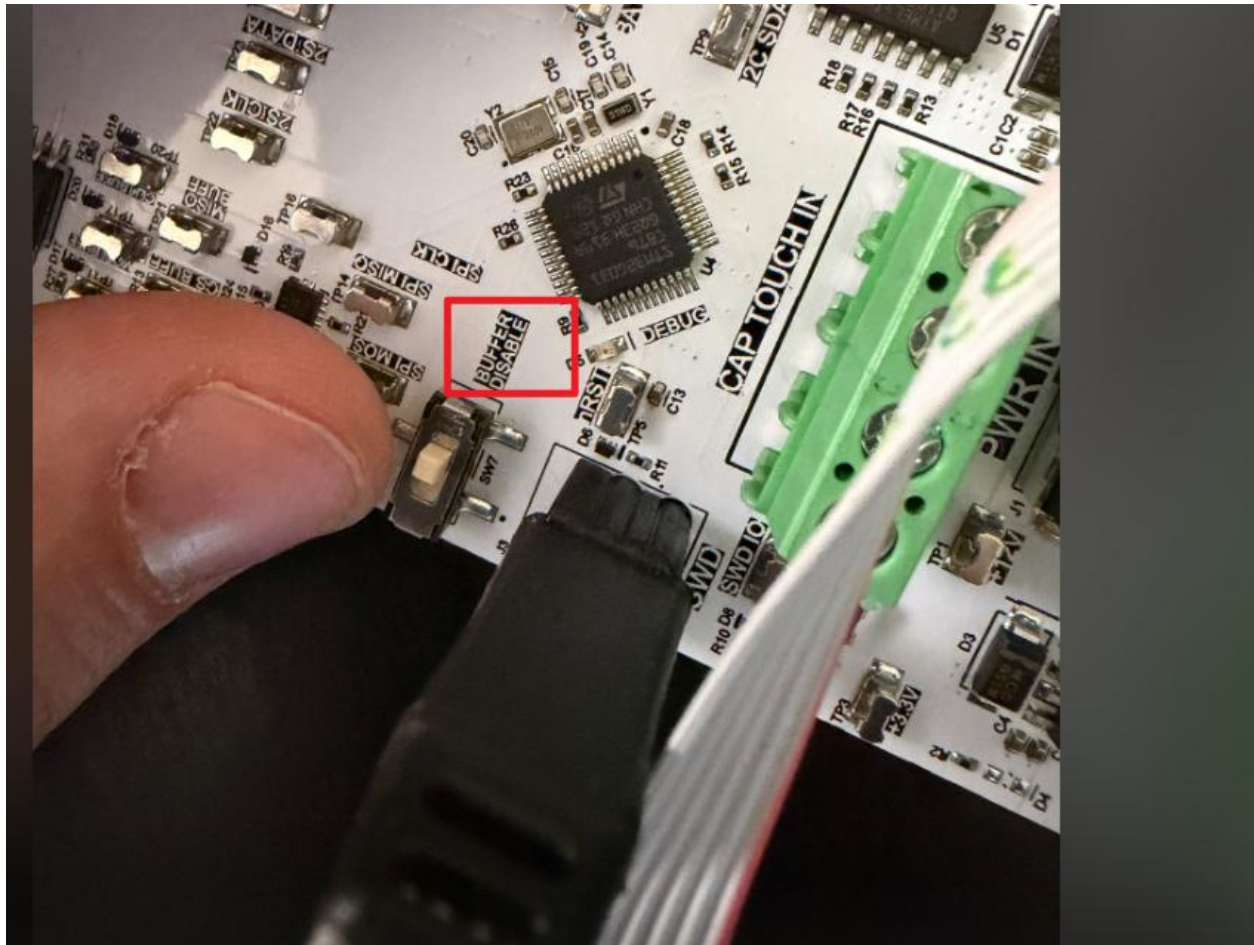
Unplug the Brain PCB’s power, remove the STLink. **The Brain is now programmed.**

To remove the programming cable, squeeze the plastic legs again and pull it out of the Brain.

Audio Memory Programming Steps

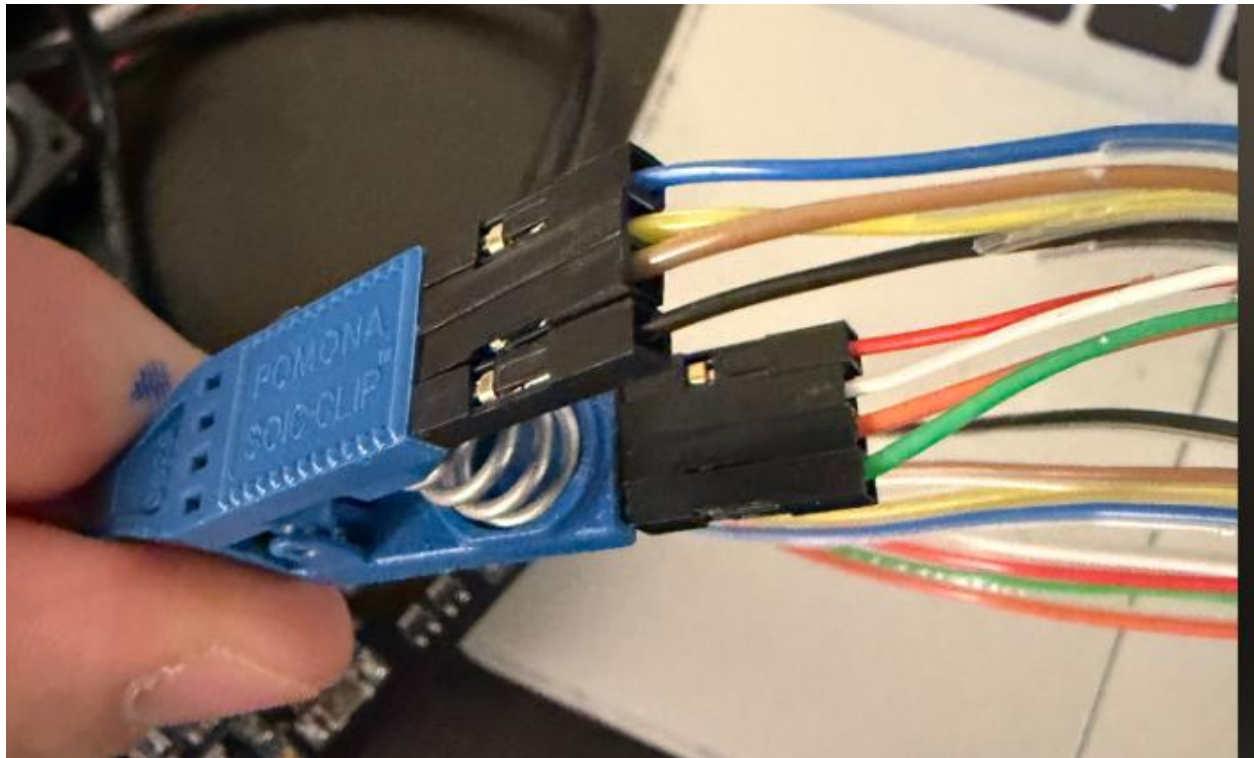
1. Disable the buffers on the Beethoven Clock Brain PCB

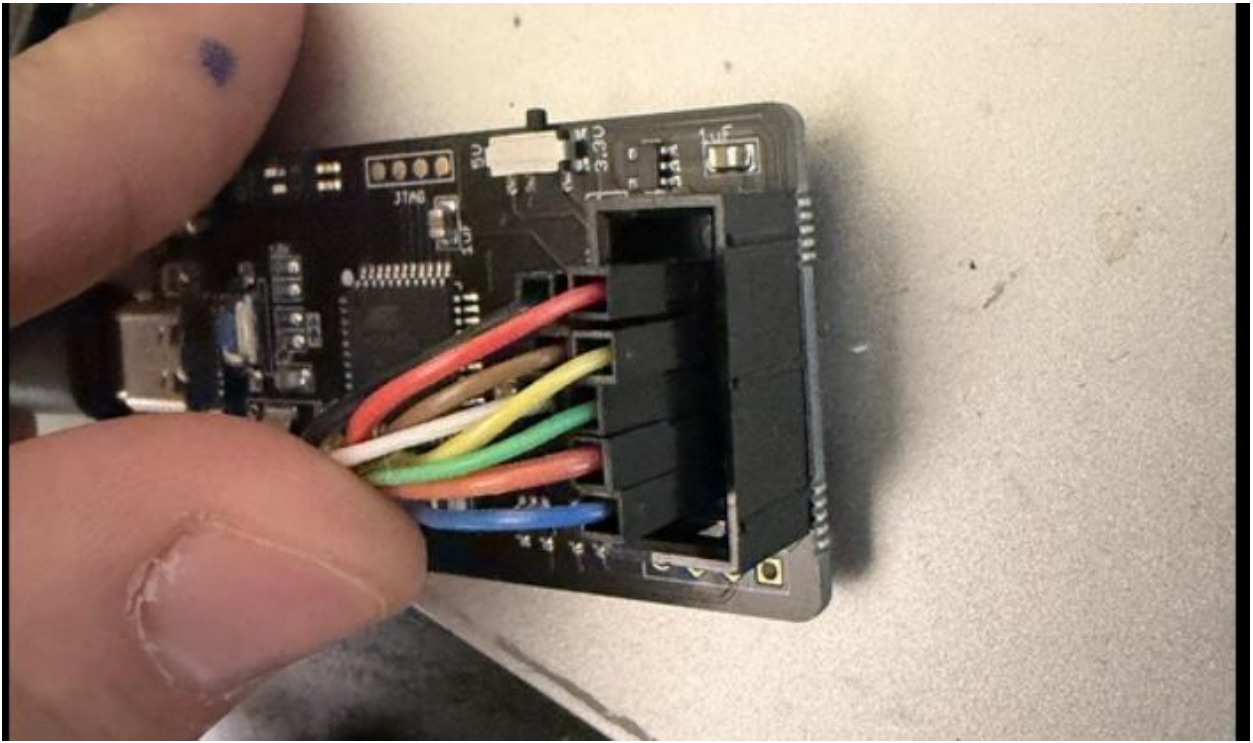
Depower the Brain PCB. Slide the white switch towards the “BUFFER DISABLE” label. This will stop the microcontroller from being able to read the memory.



2. Attach the SOIC clip to the FlashCat device

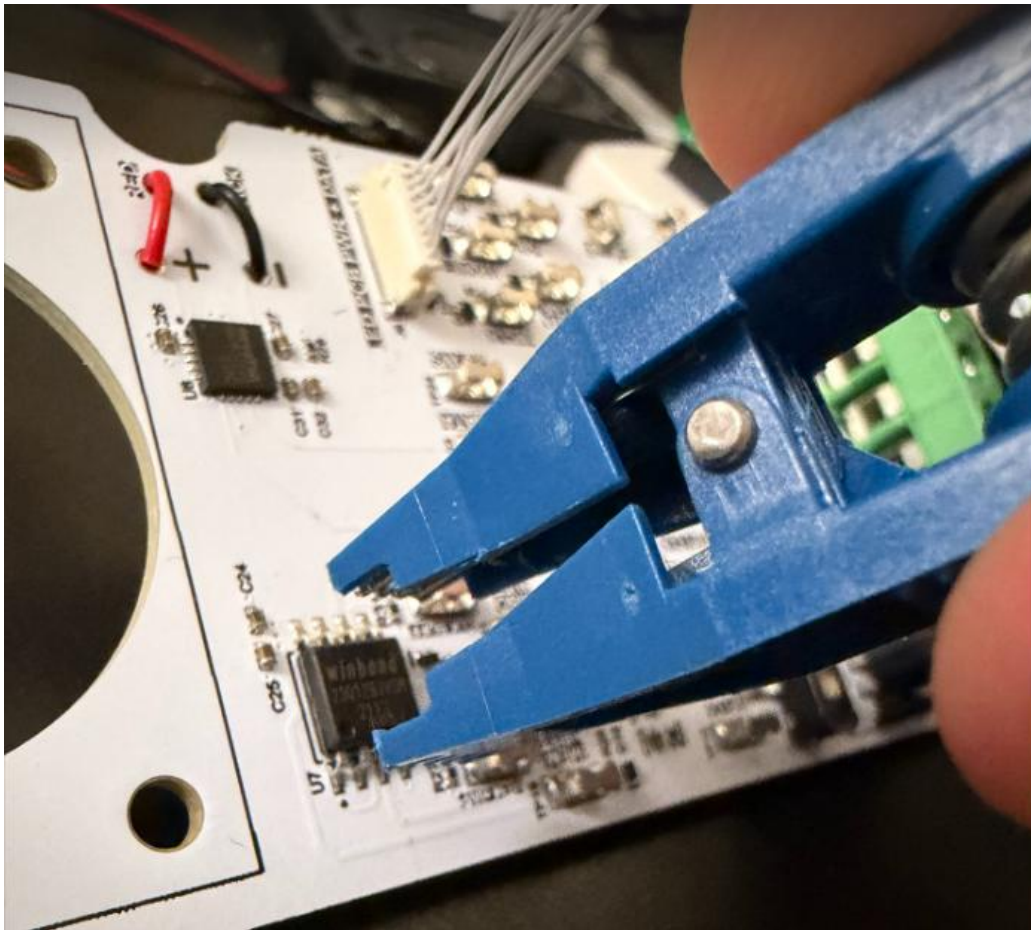
Attach the following pins of the FlashCat device to the pins of the SOIC clip (color-coded). I have given you this already done, but in case a wire comes loose, this will show you where to place it.



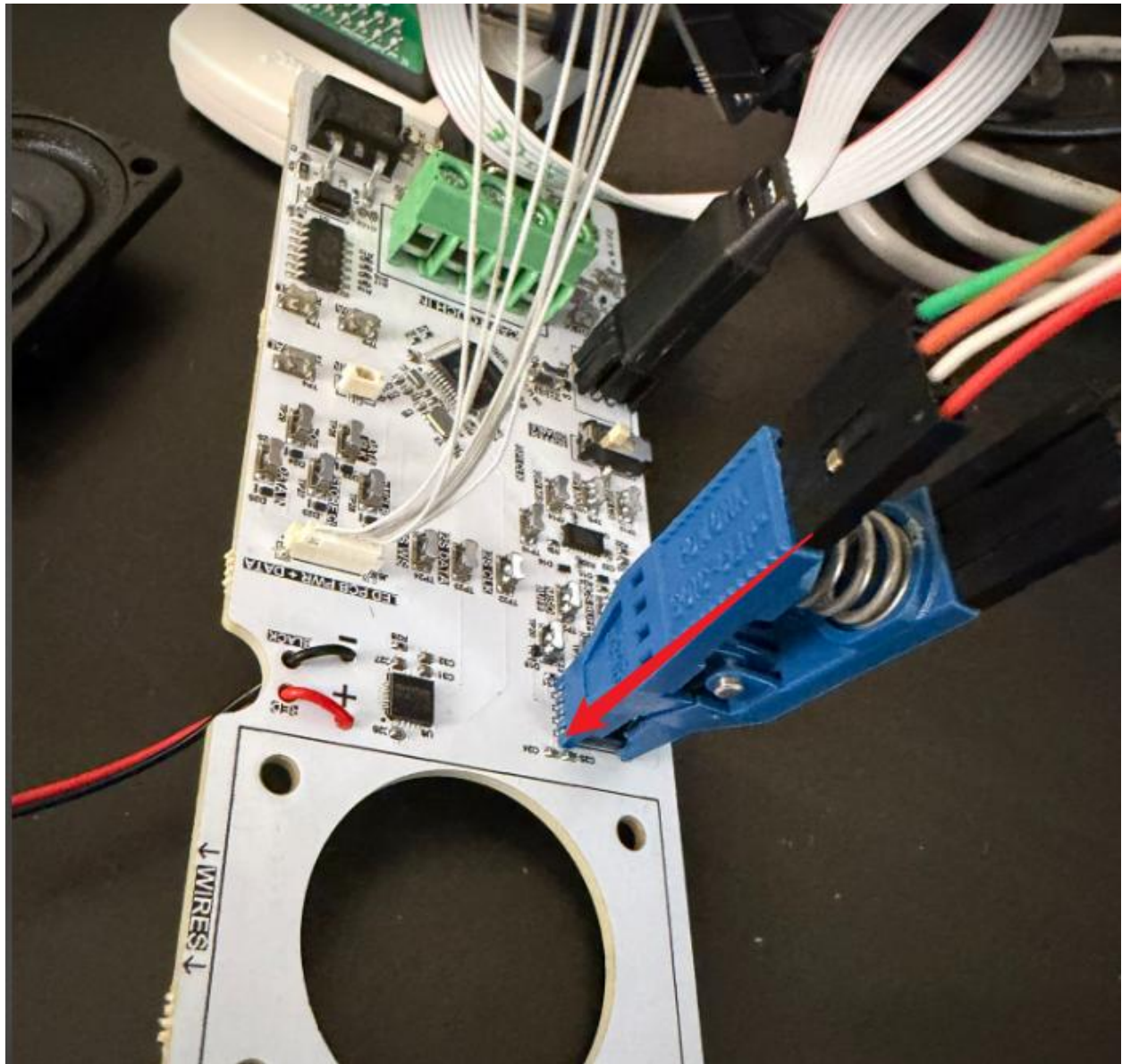


3. Attach the SOIC clip to the Memory Chip

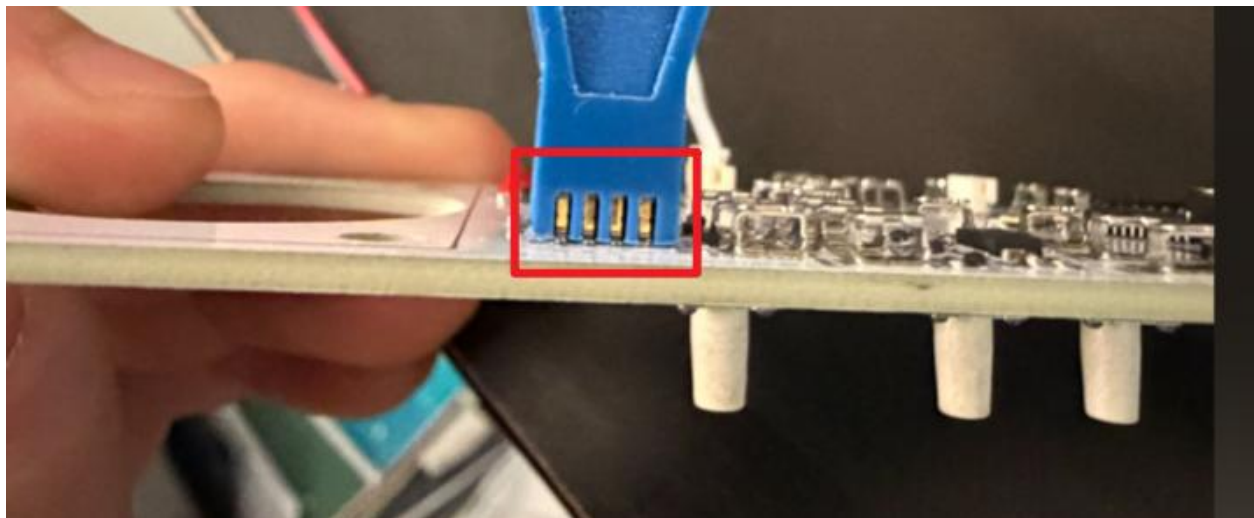
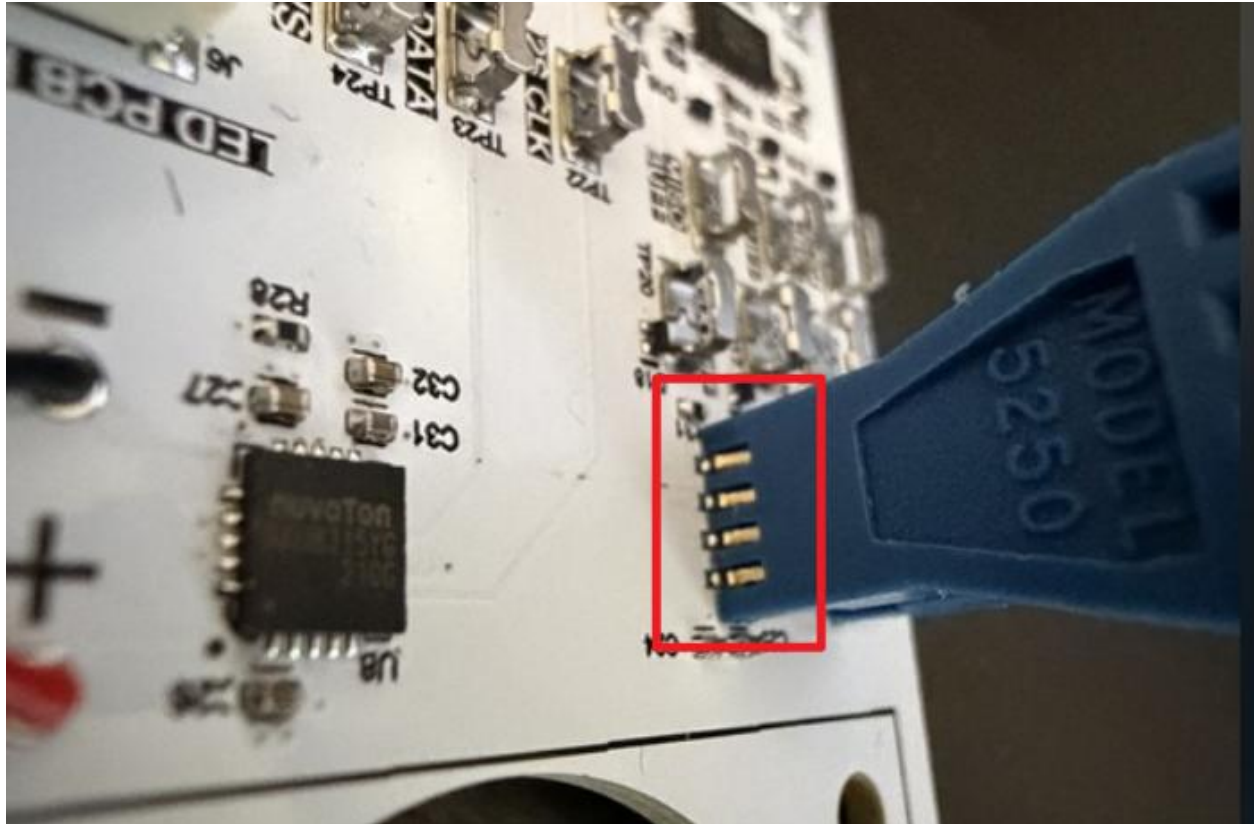
Grab the memory chip on the Beethoven Clock Brain PCB with the SOIC clip.



IMPORTANT: the side of the SOIC clip with the red, white, orange, and green wires must line up with the side of the memory chip closer to the speaker. Grabbing the memory chip the opposite way may damage it.



Check from the side that the gold contacts in the SOIC clip are aligned with the silver legs of the memory chip. If not, reposition, as awry contacts could damage the board.



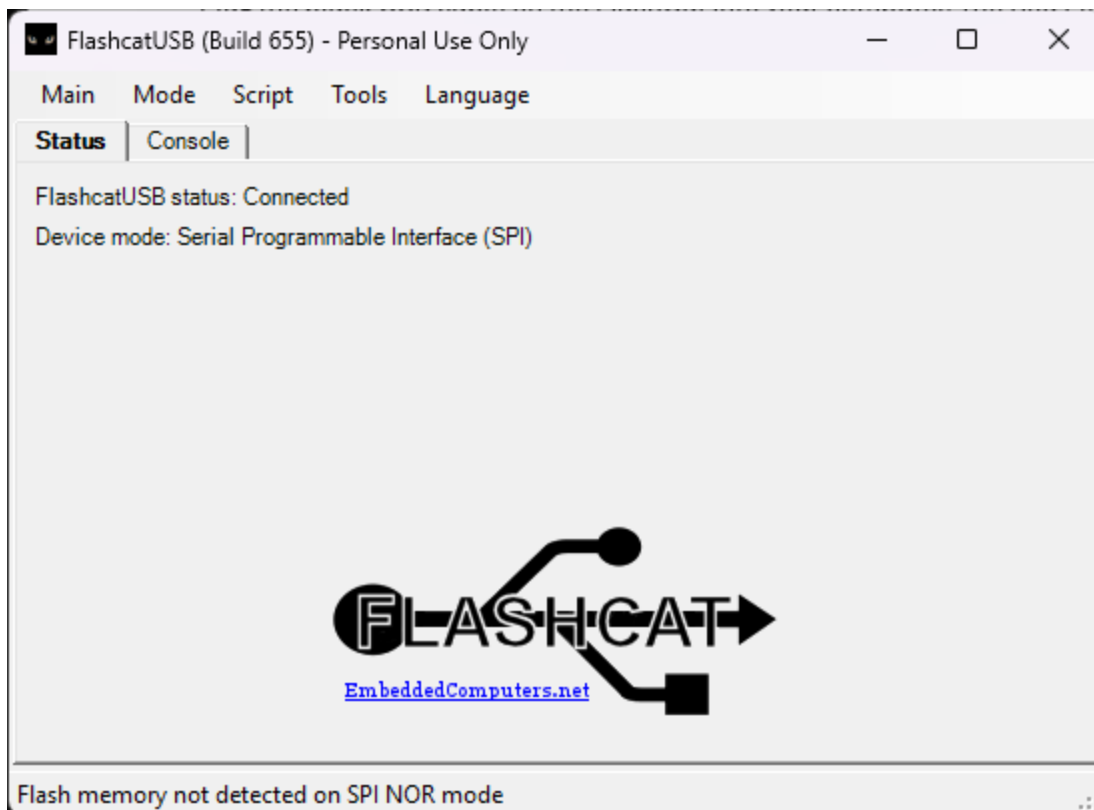
4. Plug the FlashCat into your Computer

Plug the black USB cable on the FlashCat into your computer. The Red Power LED will illuminate.

5. Open FlashcatUSB Program on your Computer

First time doing this: Download the software from [here](#). I believe you use a Mac, so be sure to download the Mac version. Note that I run windows, so I'm not sure what's involved with doing so on a Mac. This may be a question for the folks at EmbeddedComputers. It may be easier to do on a Windows device.

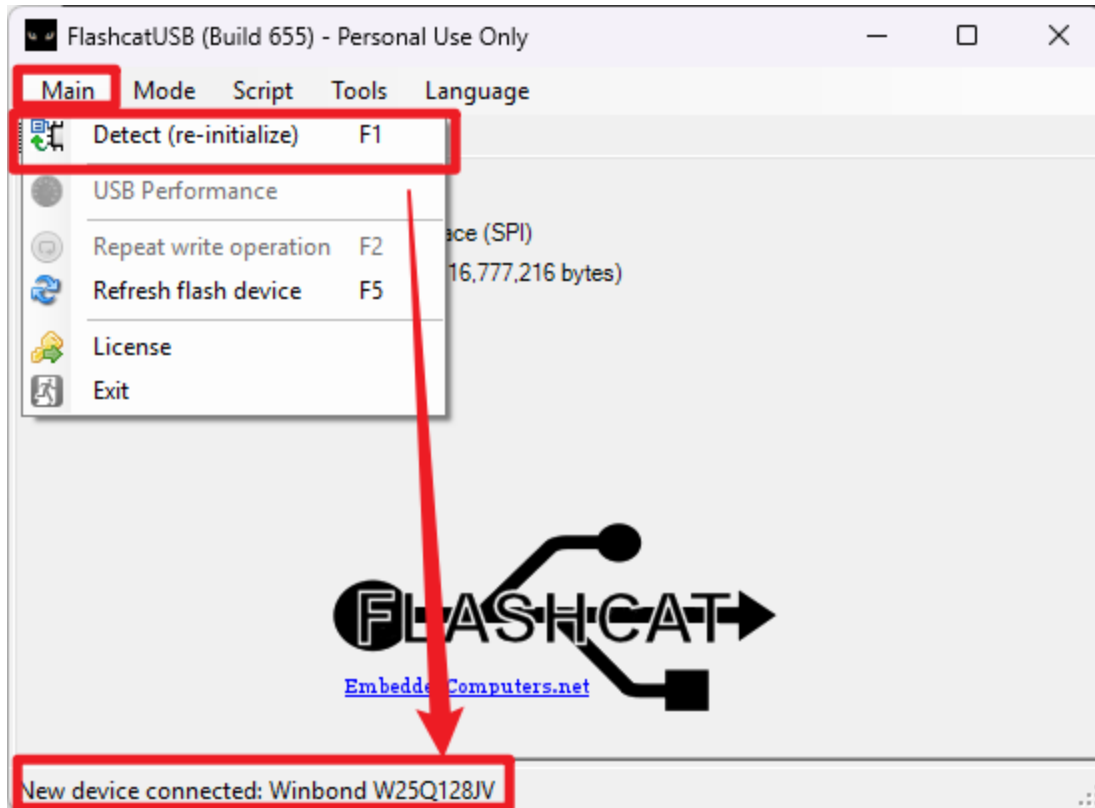
You'll be met with this screen and the status LED on the FlashCat will turn blue:



6. Power on the Beethoven Clock Brain PCB, Refresh the device list

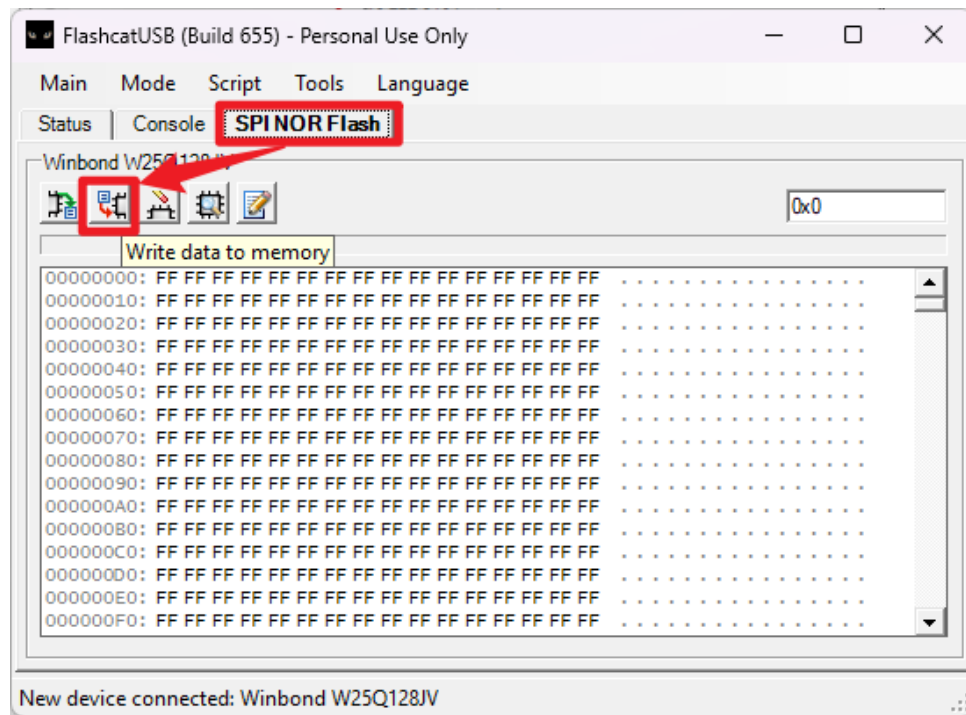
Plug the power cable into the Beethoven Clock Brain PCB.

Then, in FlashcatUSB, click Main → Detect (re-initialize). FlashCat will look for the flash chip. If everything is connected correctly, it will detect and show the device at the bottom of the screen.

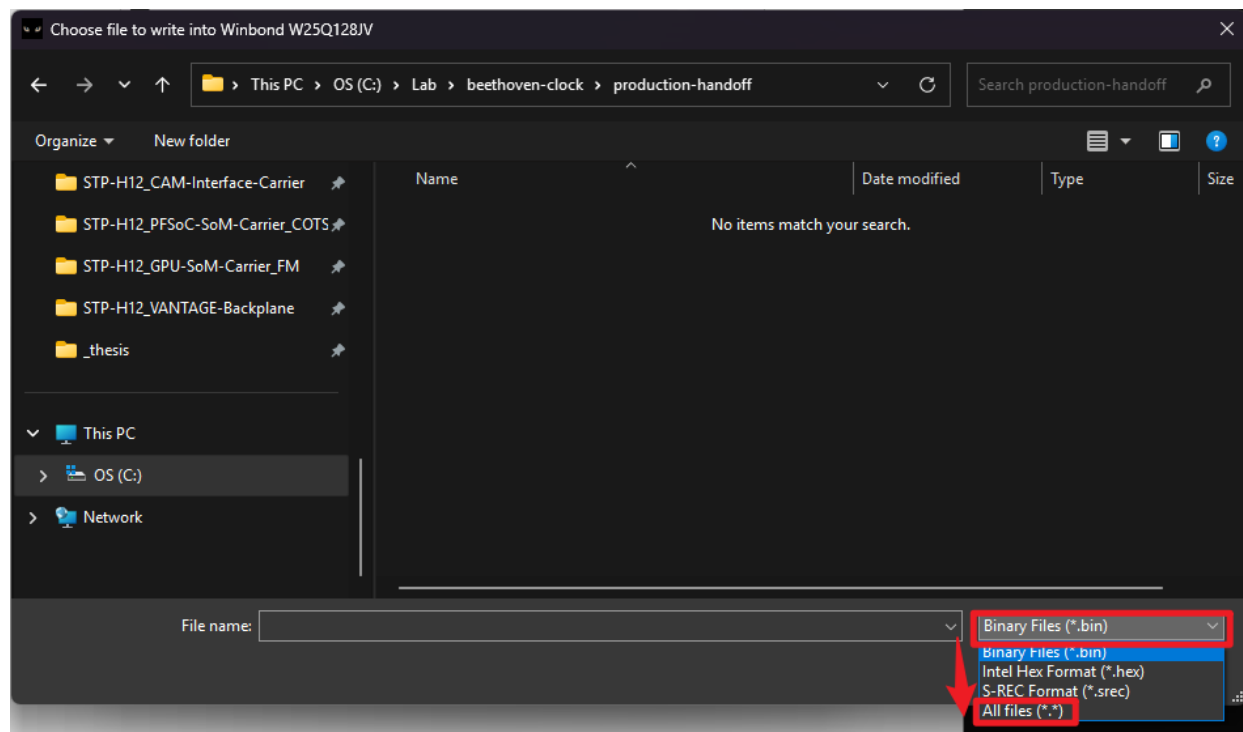


7. Tell Flashcat which Audio File to Program

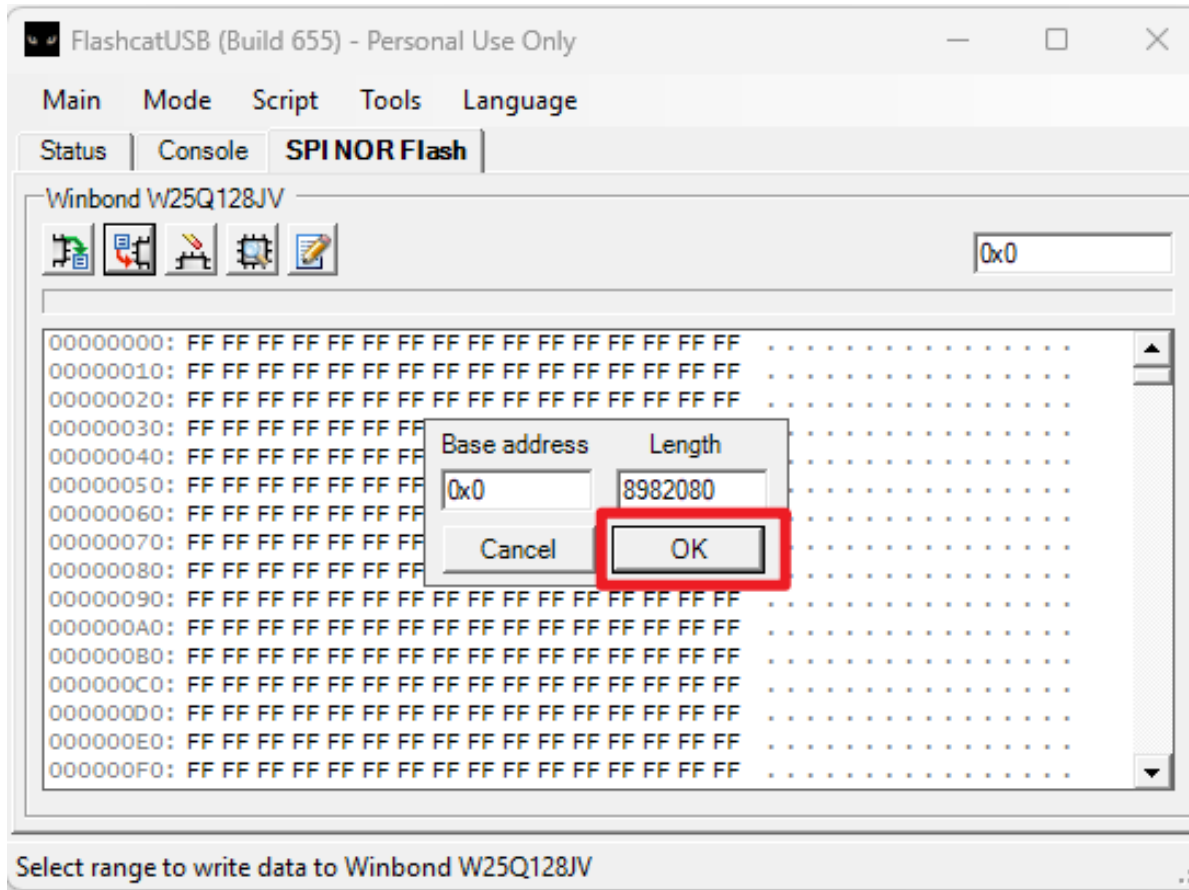
Click “SPI NOR Flash” and then click the red arrow indicating writing a file to a chip.



Navigate to the most up-to-date Beethoven clock file. It will want to look for .bin files. On Windows, the way to make it see the .wav file is to click the file type and search for “All Files.”



Then click the most up-to-date Beethoven Clock file. Select it and then click OK in the FlashCat program.



The FlashCat will start to blink and there's a progress bar at the top. This takes about 2 minutes on my machine.

8. *Power on the Beethoven Clock Brain PCB, Refresh the device list*

The memory chip is now programmed. Depower the Beethoven Clock Brain PCB, unplug the Flashcat from your computer, remove the SOIC clip.

Make sure to change the BUFFER DISABLE switch back to the side closest to the board's edge, or else the microcontroller will not be able to read the memory chip.